

Provable Security

Lecturer: Kathrin Hövelmanns

Lecture 10

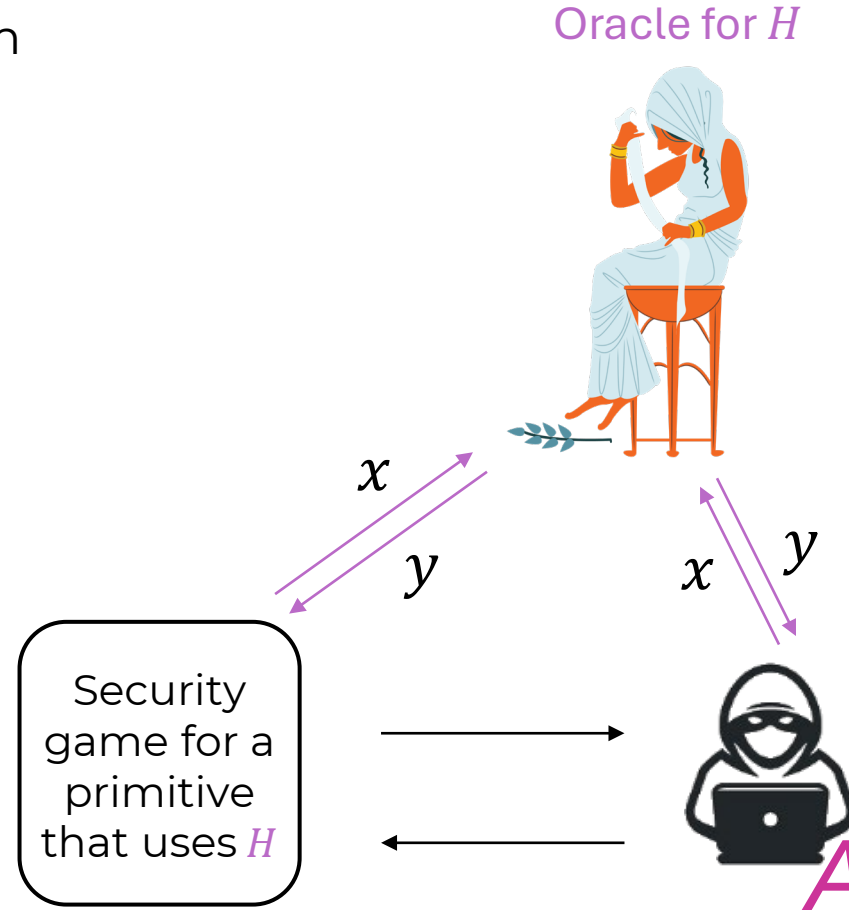
Recap

The ROM

Heuristic: Replace hash function Hash with ‘oracle box’ for a random function $H: \{0,1\}^n \rightarrow \{0,1\}^m$.

“Lazy sampling” of H :

- First-time query on x :
Pick random y from $\{0,1\}^m$
- Remember for future queries:
 $H(x) = y$
How? Store (x,y) in a table!

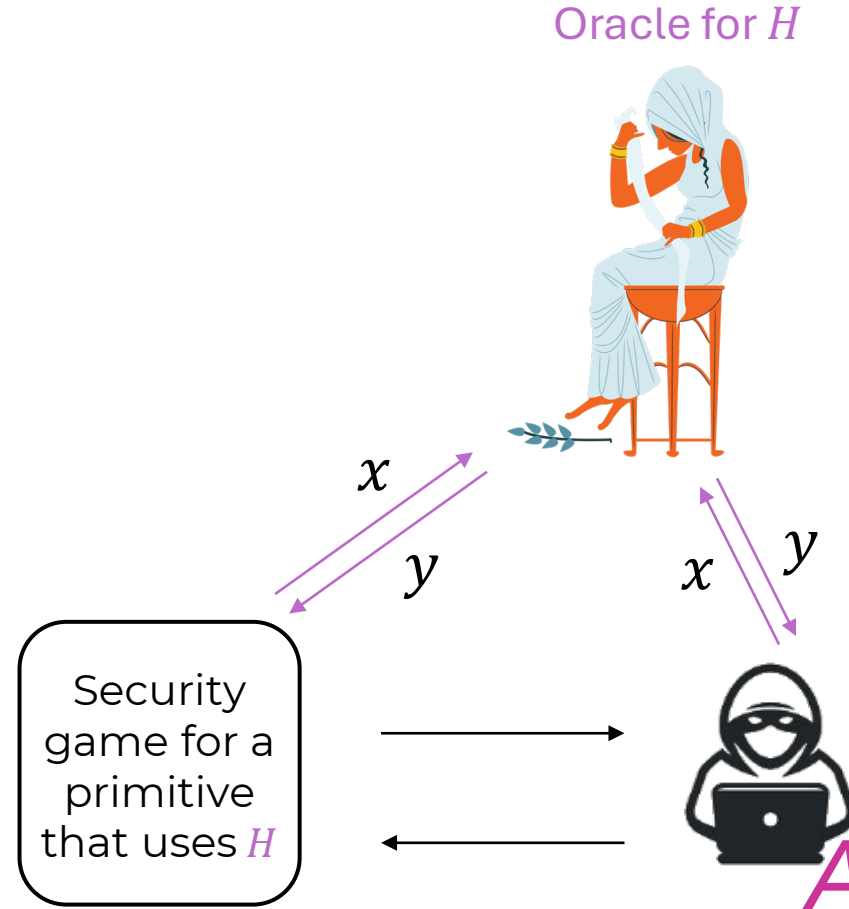


Recap

The ROM

Perks of the ROM:

- Unpredictability of values $H(x)$
- “See how **A** ticks”
- “Trick **A**”:
If we pick the answers smartly,
we can enable reductions:
 - trick **A** into solving the problem on which we want to base security
 - feign secret knowledge needed for **A**’s security game



Recap

Hashed ElGamal

Definition Hashed ElGamal

Let $\mathbb{G} = \langle g \rangle$ of prime order p . We use multiplicative notation for the group operation.

Hashed ElGamal Encryption works as follows:

$\mathcal{M} = \mathbb{G}$ $\mathcal{C} = \mathbb{G} \times \mathbb{G}$	<u>KeyGen:</u> $sk := a \leftarrow \mathbb{Z}_p$ $pk := A := g^a$ return (sk, pk)	<u>Enc($pk = A, m$):</u> $b \leftarrow \mathbb{Z}_p$ $B := g^b$ $K := Hash(A^b)$ $X := m \cdot K$ return $c := (B, X)$	<u>Dec($sk = a, c = (B, X)$):</u> $K := Hash(B^a)$ return $X \cdot K^{-1}$
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- 1.) Establish a pre-key g^{ab} like Bob would in DHKA.
- 2.) Hash the pre-key.
- 3.) Use the group-OTP to encrypt m with $K = Hash(g^{ab})$.

Recap

Hashed ElGamal

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IND-CPA security: In the ROM, breaking HEG implies finding the DH secret.

Reduction: Given CDH challenge (g, g^a, g^b) , R sets $pk := g^a$ and challenge $c := (g^b, m \cdot z)$ with random z . To extract g^{ab} , R picks a random adversary query to the RO modelling *Hash*.

Recap

IND-CCA secure PKE: DHIES

Definition DHIES

Let $\mathbb{G} = \langle g \rangle$ of prime order p . We use multiplicative notation for the group operation. Let $\text{MAC} = (\text{Tag}, \text{Ver})$ be a MAC.

DHIES Encryption works as follows:

$\mathcal{M} = \mathbb{G}$ $\mathcal{C} = \mathbb{G} \times \mathbb{G}$	<u>KeyGen:</u> $sk := a \leftarrow \mathbb{Z}_p$ $pk := A := g^a$ return (sk, pk)	<u>Enc($pk = A, m$):</u> $K := \text{Hash}(A^b)$ $K_E K_M := K$ $X := m \cdot K_E$ $t := \text{Tag}(K_M, X)$ return $c := (B, X, t)$	<u>Dec($sk = a, c = (B, X, t)$):</u> $K_E K_M := \text{Hash}(B^a)$ If (X, t) verifies under K_M : return $X \cdot K_E^{-1}$ Else reject
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- 1.) Establish a pre-key K like in HEG.
- 2.) Split K into one key for the group-OTP and one for the MAC.
- 3.) Use the group-OTP to encrypt m with K_E .
- 4.) When decrypting, punish cts with “invalid” tags by rejecting.

Yet another transformation?

New goal: Generic IND-CCA secure KEM designs

So far, we saw:

- how to get CCA-secure PKE from DH-style assumptions.

But:

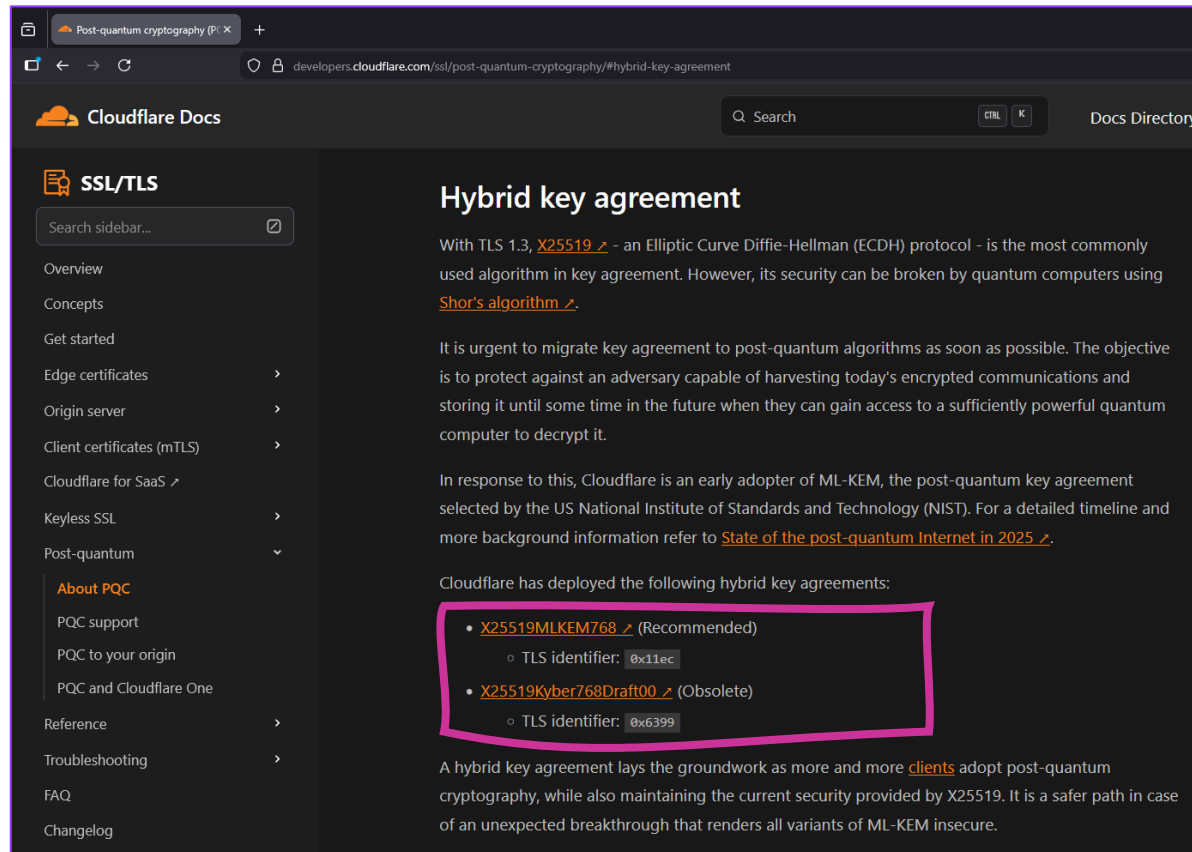
- Many protocols don't actually need full PKE:
 - We only need ways to establish symmetric keys (“**K**ey **E**ncapsulation **M**echanisms”).
 - Many protocols do:
 - KEM → shared key
 - Then AEAD
- Plus: DH-based designs are not quantum-safe

New goal: How to build CCA-secure **KEMs**, **generically**?

- We now see a recipe called **F**ujisaki–**O**kamoto transform.
- Many recently established pq KEMs (e.g., NIST standard ML-KEM) use FO.

Yet another transformation?

New goal: Generic IND-CCA secure KEM designs



The screenshot shows the Cloudflare Docs website. The left sidebar is titled "SSL/TLS" and contains a search bar and a list of navigation links: Overview, Concepts, Get started, Edge certificates, Origin server, Client certificates (mTLS), Cloudflare for SaaS, Keyless SSL, Post-quantum, About PQC, PQC support, PQC to your origin, PQC and Cloudflare One, Reference, Troubleshooting, FAQ, and Changelog. The main content area is titled "Hybrid key agreement". It contains the following text:

With TLS 1.3, [X25519](#) - an Elliptic Curve Diffie-Hellman (ECDH) protocol - is the most commonly used algorithm in key agreement. However, its security can be broken by quantum computers using [Shor's algorithm](#).

It is urgent to migrate key agreement to post-quantum algorithms as soon as possible. The objective is to protect against an adversary capable of harvesting today's encrypted communications and storing it until some time in the future when they can gain access to a sufficiently powerful quantum computer to decrypt it.

In response to this, Cloudflare is an early adopter of ML-KEM, the post-quantum key agreement selected by the US National Institute of Standards and Technology (NIST). For a detailed timeline and more background information refer to [State of the post-quantum Internet in 2025](#).

Cloudflare has deployed the following hybrid key agreements:

- [X25519MLKEM768](#) (Recommended)
 - TLS identifier: `0x11ec`
- [X25519Kyber768Draft00](#) (Obsolete)
 - TLS identifier: `0x6399`

A hybrid key agreement lays the groundwork as more and more [clients](#) adopt post-quantum cryptography, while also maintaining the current security provided by X25519. It is a safer path in case of an unexpected breakthrough that renders all variants of ML-KEM insecure.

Yet another transformation?

References on IND-CCA secure designs and KEMs

PKE transforms:

Original FO, 1999: E. Fujisaki, T. Okamoto. *Secure integration of asymmetric and symmetric encryption schemes*. (Journal version in 2013)

REACT, 2001: T. Okamoto, D. Pointcheval. *REACT: Rapid Enhanced-security Asymmetric Cryptosystem Transform*.

Hashed ElGamal/DIES, 2001: M. Abdalla, M. Bellare, P. Rogaway. *The oracle Diffie-Hellman assumptions and an analysis of DHIES*.

GEM, 2002: J. Coron, H. Handschuh, M. Joye, P. Paillier, D. Pointcheval, C. Tymen. *GEM: A generic chosen-ciphertext secure encryption method*.

Cramer-Shoup, 2003: R. Cramer, V. Shoup. *Design and analysis of practical public-key encryption schemes secure against adaptive chosen ciphertext attack*.

KEMs:

Initial KEM analysis, 2003: A. W. Dent. *A designer's guide to KEMs*.

FO for pq KEMs, 2017: D. Hofheinz, K. Hövelmanns, E. Kiltz. *A Modular Analysis of the Fujisaki–Okamoto Transformation*.

What is a KEM?

A definition that abstracts “establish a symmetric key”

KEM = Public-key transport of symmetric keys:

1. **KeyGen**: Create public-key pair pk, sk



Alice



Bob



What is a KEM?

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1. **KeyGen**: Create public-key pair pk, sk
2. **Encapsulate**: Creates key K and ciphertext c that ‘encrypts’ K

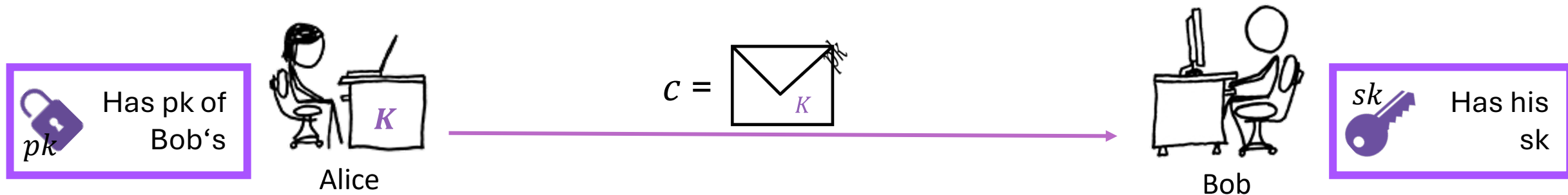


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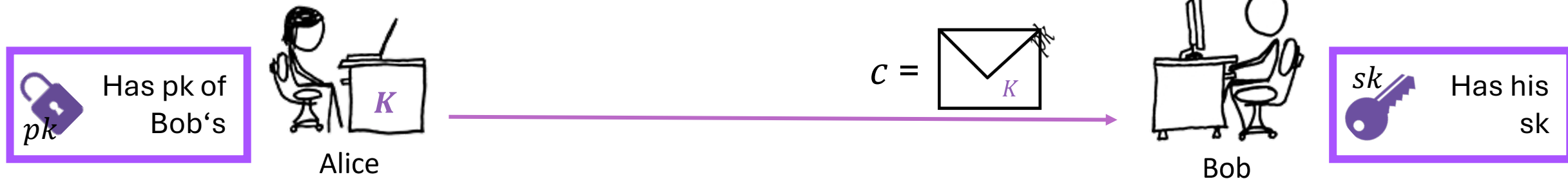


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1. **KeyGen**: Create public-key pair pk, sk
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3. **Decapsulate**(sk, c): ‘Decrypts’ c into K

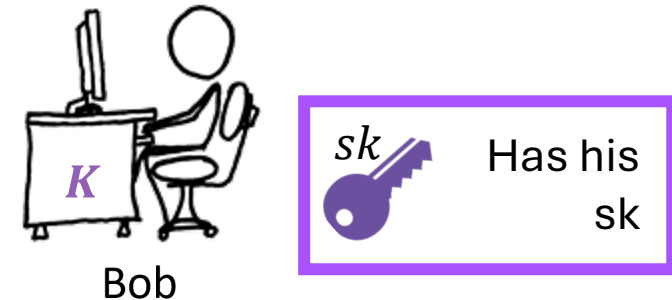
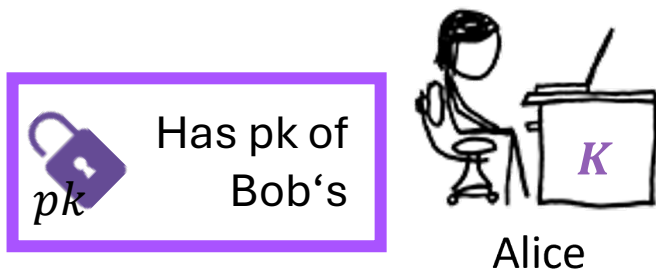


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3. **Decapsulate**(sk, c): ‘Decrypts’ c into K



What is a KEM?

A definition that abstracts “establish a symmetric key”

Definition Key Encapsulation Mechanism

A KEM consists of the following algorithms:

- KeyGen: algorithm that outputs a key pair (pk, sk) .
- Encaps: algorithm that inputs a public key pk and outputs a ciphertext $c \in \mathcal{C}$ and a key $K \in \mathcal{K}$.
- Decaps: deterministic algorithm that inputs a secret key sk and ciphertext $c \in \mathcal{C}$, and outputs a key $K \in \mathcal{K}$.

We call the set $\mathcal{K}$ of possible keys the scheme's **key space**, and $\mathcal{C}$ the **ciphertext space**.

What is a KEM?

“Weak” security definition for KEMs: IND-CPA

IND-CPA captures that output keys of KEM look random:

Definition IND-CPA security

We define the **IND-CPA advantage** of an adversary $\mathcal{A}$ against KEM as

$$\text{Adv}_{\text{KEM}}^{\text{IND-CPA}}(\mathcal{A}) := \left| \Pr[\text{IND-CPA}^{\mathcal{A}} \Rightarrow 1] - \frac{1}{2} \right|.$$

Game $\text{IND-CPA}_{\text{KEM}}^{\mathcal{A}}$

Initialize:

$(pk, sk) \leftarrow \text{KeyGen}$

$b \stackrel{\$}{\leftarrow} \{0,1\}$

$(K_0^*, c^*) \leftarrow \text{Encaps}(pk)$

$K_1^* \stackrel{\$}{\leftarrow} \mathcal{K}$

return pk, c^*, K_b^*

Finalize ($b' \in \{0,1\}$):

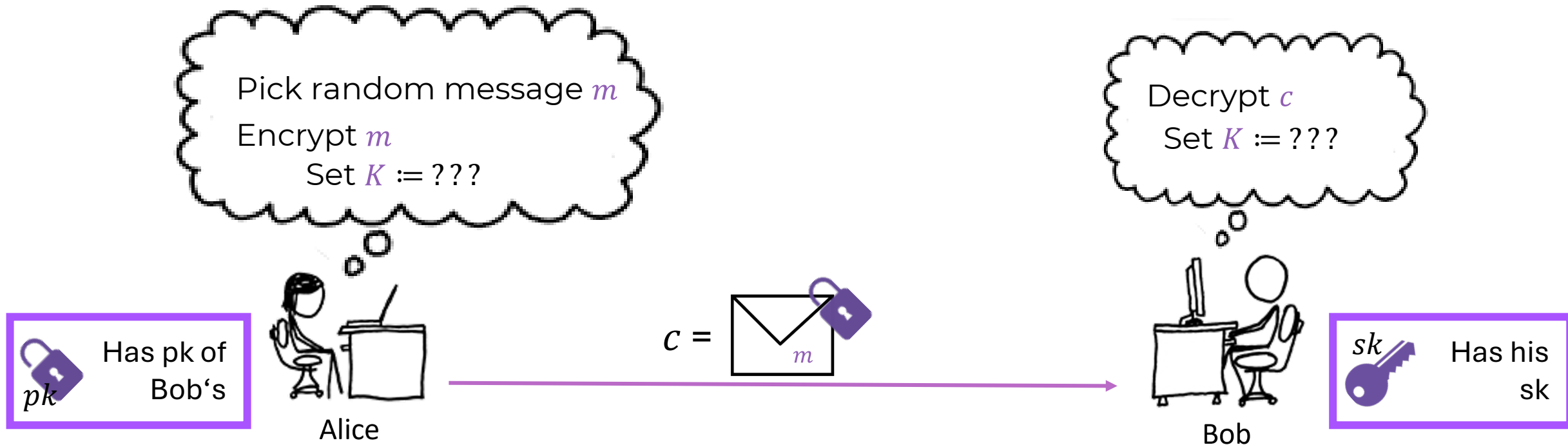
return $b' =_? b$

- Victim encapsulates
- $\mathcal{A}$ sees c^* and key that either matches c^* or is a random one
- $\mathcal{A}$ guesses: “real key” or “random key”

Fujisaki-Okamoto

Step 1: U-transform - OW PKE to IND-CPA KEM

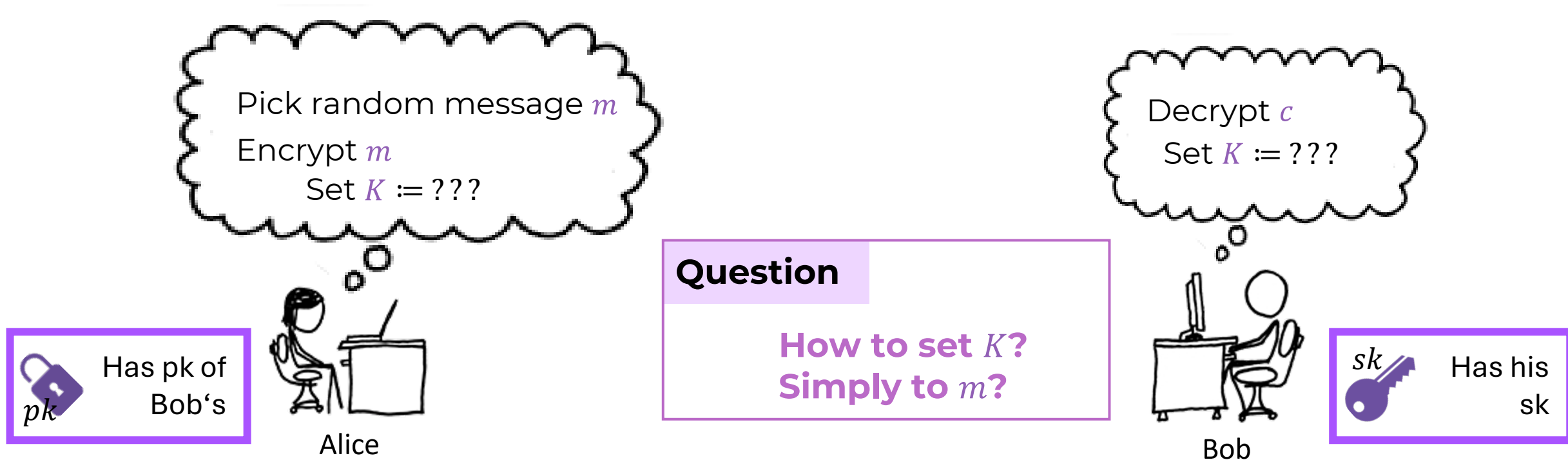
Goal: Use a PKE scheme to securely share a key K .



Fujisaki-Okamoto

Step 1: U-transform - OW PKE to IND-CPA KEM

Goal: Use a PKE scheme to securely share a key K .



Fujisaki-Okamoto

Step 1: U-transform - OW PKE to IND-CPA KEM

Goal: Use a PKE scheme to securely share a key K .

Higher-level protocols usually need K as n -bitstrings (e.g. $n = 128/192/256$). m might be a group element, a bitstring of the wrong length, ...

K should look random, and m can have a recognizable structure (algebraic, encoding, ...) that would carry to the protocol layer.

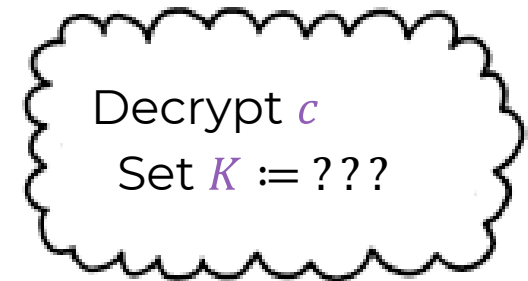
If PKE is deterministic, the KEM is not IND-CPA secure!



Alice

Question

How to set K ?
Simply to m ?



Bob



Fujisaki-Okamoto

Step 1: U-transform - OW PKE to IND-CPA KEM

Goal: Use a PKE scheme to securely share a key K .



Fujisaki-Okamoto

Step 1: U-transform - OW PKE to IND-CPA KEM

Definition FO transform, step 1: U-transform

Let $\text{PKE} = (\text{KeyGen}, \text{Enc}, \text{Dec})$ be a PKE scheme with message space $\mathcal{M}$, and let $H: \{0,1\}^* \rightarrow \{0,1\}^n$ be a hash function.

We define KEM with key space $\mathcal{K} = \{0,1\}^n$ and algorithms $\text{KeyGen}_{\text{KEM}} := \text{KeyGen}$ and

<u>Encaps(pk):</u> $m \xleftarrow{\$} \mathcal{M}$ $c \leftarrow \text{Enc}(\text{pk}, m)$ $K := H(m, c)$ return (K, c)	<u>Decaps(sk, c):</u> $m' := \text{Dec}(\text{sk}, c)$ if $m' = \perp$ return $\perp$ else return $K := H(m', c)$
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In the literature, this PKE-to-KEM transformation is called **$U^\perp$ transform**, and the resulting KEM is denoted as $\text{KEM}^\perp$ or $U^\perp[\text{PKE}, H]$.

Fujisaki-Okamoto

Step 1: U-transform - OW PKE to IND-CPA KEM

We formalize the security property of PKE on which we will base the security of KEM[⊥]: **One-Way** security.

Definition OW security

We define the **OW advantage** of an adversary $\mathcal{A}$ against PKE as

$$\text{Adv}_{\text{PKE}}^{\text{OW}}(\mathcal{A}) := \Pr[\text{OW}_{\text{PKE}}^{\mathcal{A}} \Rightarrow 1].$$

Game $\text{OW}_{\text{PKE}}^{\mathcal{A}}$

Initialize:

$(pk, sk) \leftarrow \text{KeyGen}$

$m^* \xleftarrow{\$} \mathcal{M}$

$c^* \leftarrow \text{Enc}(pk, m^*)$

return pk, c^*

Finalize ($m' \in \mathcal{M}$):

return $m' =_? m^*$

Fujisaki-Okamoto

Step 1: U-transform - OW PKE to IND-CPA KEM

Security claim (PKE OW $\Rightarrow$ KEM[⊥] IND-CPA)

For any IND-CPA adversary $\mathcal{A}$ against KEM[⊥], issuing at most q_H queries to random oracle H , there exists an OW adversary $\mathcal{B}$ against PKE such that

$$\text{Adv}_{\text{KEM}^\perp}^{\text{IND-CPA}}(\mathcal{A}) \leq q_H \cdot \text{Adv}_{\text{PKE}}^{\text{OW}}(\mathcal{B}).$$

$\mathcal{B}$ runs in about the same time as $\mathcal{A}$.

Proof intuition:

In the ROM, $\mathcal{A}$ can't tell $K_0^* := H(m^*, c^*)$ from $K_1^* \xleftarrow{\$} \mathcal{K}$ without querying H on m^* .

Which implies knowing m^* .

Which implies breaking OW security of PKE.

Game IND-CPA ^{$\mathcal{A}$} _{KEM[⊥]}

Initialize:

$(pk, sk) \leftarrow \text{KeyGen}$

$b \xleftarrow{\$} \{0,1\}$

// $(K_0^*, c^*) \leftarrow \text{Encaps}(pk)$:

$m^* \xleftarrow{\$} \mathcal{M}$

$c^* \leftarrow \text{Enc}(pk, m^*)$

$K_0^* := H(m^*, c^*)$

$K_1^* \xleftarrow{\$} \mathcal{K}$

return pk, c^*, K_b^*

$H(m, c)$:

if $\exists K$ such that

$(m, c, K) \in T_H$

return K

$K \xleftarrow{\$} \mathcal{K}$

$T_H := T_H \cup \{(m, c, K)\}$

return K

Finalize ($b' \in \{0,1\}$):

return $b' =_? b$

Fujisaki-Okamoto

Step 1: U-transform - OW PKE to IND-CPA KEM

Security claim (PKE OW $\Rightarrow$ KEM[⊥] IND-CPA)

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$\mathcal{B}$ runs in about the same time as $\mathcal{A}$.

Proof intuition:

$K_0 = H(m^*, c^*)$ looks fully random unless $\mathcal{A}$ queries H on (m^*, c^*) .

Cover this via a **single game-hop: raise flag CHALL and abort** if H is ever queried on (m^*, c^*) .

Then use this event to attack PKE: OW solution m^* can be found by monitoring RO H .

Game IND-CPA ^{$\mathcal{A}$} _{KEM[⊥]}

Initialize:

$(pk, sk) \leftarrow \text{KeyGen}$

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return K

Finalize ($b' \in \{0,1\}$):

return $b' =_? b$

Fujisaki-Okamoto

Step 1: U-transform - OW PKE to IND-CPA KEM

Security claim (PKE OW $\Rightarrow$ KEM[⊥] IND-CPA)

For any IND-CPA adversary $\mathcal{A}$ against KEM[⊥], issuing at most q_H queries to random oracle H , there exists an OW adversary $\mathcal{B}$ against PKE such that

$$\text{Adv}_{\text{KEM}^\perp}^{\text{IND-CPA}}(\mathcal{A}) \leq q_H \cdot \text{Adv}_{\text{PKE}}^{\text{OW}}(\mathcal{B}).$$

$\mathcal{B}$ runs in about the same time as $\mathcal{A}$.

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Cover this via a **single game-hop: raise flag CHALL and abort** if H is ever queried on (m^*, c^*) .

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Games G_0 and G_1

Initialize:

$(pk, sk) \leftarrow \text{KeyGen}$

$b \xleftarrow{\$} \{0,1\}$

$// (K_0^*, c^*) \leftarrow \text{Encaps}(pk):$

$m^* \xleftarrow{\$} \mathcal{M}$

$c^* \leftarrow \text{Enc}(pk, m^*)$

$K_0^* := H(m^*, c^*)$

$K_1^* \xleftarrow{\$} \mathcal{K}$

return pk, c^*, K_b^*

Finalize ($b' \in \{0,1\}$):

return $b' =_? b$

$H(m, c):$

if $(m, c) = (m^*, c^*)$
and query was by $\mathcal{A}$

CHALL := true

Abort

if $\exists K$ such that

$(m, c, K) \in T_H$

return K

$K \xleftarrow{\$} \mathcal{K}$

$T_H := T_H \cup \{(m, c, K)\}$

return K

Fujisaki-Okamoto

Step 1: U-transform - OW PKE to IND-CPA KEM

Security claim (PKE OW $\Rightarrow$ KEM $^\perp$ IND-CPA)

For any IND-CPA adversary $\mathcal{A}$ against KEM $^\perp$, issuing at most q_H queries to random oracle H , there exists an OW adversary $\mathcal{B}$ against PKE such that

$$\text{Adv}_{\text{KEM}^\perp}^{\text{IND-CPA}}(\mathcal{A}) \leq q_H \cdot \text{Adv}_{\text{PKE}}^{\text{OW}}(\mathcal{B}).$$

$\mathcal{B}$ runs in about the same time as $\mathcal{A}$.

Game analysis:

G_0 = IND-CPA game for KEM $^\perp$.

$$\text{Adv}_{\text{KEM}^\perp}^{\text{IND-CPA}}(\mathcal{A}) = \left| \Pr[G_0^{\mathcal{A}} \Rightarrow 1] - \frac{1}{2} \right|.$$

Games G_0 and G_1

Initialize:

$(pk, sk) \leftarrow \text{KeyGen}$

$b \xleftarrow{\$} \{0,1\}$

// $(K_0^*, c^*) \leftarrow \text{Encaps}(pk)$:

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$c^* \leftarrow \text{Enc}(pk, m^*)$

$K_0^* := H(m^*, c^*)$

$K_1^* \xleftarrow{\$} \mathcal{K}$

return pk, c^*, K_b^*

Finalize ($b' \in \{0,1\}$):

return $b' =_? b$

$H(m, c)$:

if $(m, c) = (m^*, c^*)$
and query was by $\mathcal{A}$

CHALL := true

Abort

if $\exists K$ such that

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return K

$K \xleftarrow{\$} \mathcal{K}$

$T_H := T_H \cup \{(m, c, K)\}$

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Fujisaki-Okamoto

Step 1: U-transform - OW PKE to IND-CPA KEM

Security claim (PKE OW $\Rightarrow$ KEM[⊥] IND-CPA)

For any IND-CPA adversary $\mathcal{A}$ against KEM[⊥], issuing at most q_H queries to random oracle H , there exists an OW adversary $\mathcal{B}$ against PKE such that

$$\text{Adv}_{\text{KEM}^\perp}^{\text{IND-CPA}}(\mathcal{A}) \leq q_H \cdot \text{Adv}_{\text{PKE}}^{\text{OW}}(\mathcal{B}).$$

$\mathcal{B}$ runs in about the same time as $\mathcal{A}$.

Game analysis:

G_1 : Equals G_0 unless **CHALL** happens. Due to the difference lemma,

$$|\Pr[G_0^{\mathcal{A}} \Rightarrow 1] - \Pr[G_1^{\mathcal{A}} \Rightarrow 1]| \leq \Pr[\text{CHALL}],$$

so

$$\text{Adv}_{\text{KEM}}^{\text{IND-CPA}}(\mathcal{A}) \leq \Pr[\text{CHALL}] + \left| \Pr[G_1^{\mathcal{A}} \Rightarrow 1] - \frac{1}{2} \right|.$$

Games G_0 and G_1

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Finalize ($b' \in \{0,1\}$):

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Fujisaki-Okamoto

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$\mathcal{B}$ runs in about the same time as $\mathcal{A}$.

Game analysis:

G_1 : $K_0 = H(m^*, c^*)$ is not given to $\mathcal{A}$ by the RO, so bit b is independent from $\mathcal{A}$'s view. I claim:

$$\Pr[G_1^{\mathcal{A}} \Rightarrow 1] = 1/2.$$

If that is true, then

$$\text{Adv}_{\text{KEM}}^{\text{IND-CPA}}(\mathcal{A}) \leq \Pr[\text{CHALL}].$$

Games G_0 and G_1

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$\mathcal{B}$ runs in about the same time as $\mathcal{A}$.

Game analysis:

G_1 : $K_0 = H(m^*, c^*)$ is not given to $\mathcal{A}$ by the RO, so bit b is independent from $\mathcal{A}$'s view.

We now show:

$$\Pr[G_1^{\mathcal{A}} \Rightarrow 1] = 1/2.$$

Games G_0 and G_1

Initialize:

$(pk, sk) \leftarrow \text{KeyGen}$

$b \xleftarrow{\$} \{0,1\}$

$((K_0^*, c^*) \leftarrow \text{Encaps}(pk):$

$m^* \xleftarrow{\$} \mathcal{M}$

$c^* \leftarrow \text{Enc}(pk, m^*)$

$K_0^* := H(m^*, c^*)$

$K_1^* \xleftarrow{\$} \mathcal{K}$

return pk, c^*, K_b^*

Finalize ($b' \in \{0,1\}$):

return $b' =_? b$

$H(m, c)$:

if $(m, c) = (m^*, c^*)$
and query was by $\mathcal{A}$

CHALL := true

Abort

if $\exists K$ such that

$(m, c, K) \in T_H$

return K

$K \xleftarrow{\$} \mathcal{K}$

$T_H := T_H \cup \{(m, c, K)\}$

return K

Fujisaki-Okamoto

Step 1: U-transform - OW PKE to IND-CPA KEM

We now show: Since b is independent of $\mathcal{A}$'s view,

$$\Pr[G_1^{\mathcal{A}} \Rightarrow 1] = 1/2 .$$

We rewrite G_1 : In Initialize, we inline $K_0^* := H(m^*, c^*)$.

Games G_0 and G_1

Initialize:

$(pk, sk) \leftarrow \text{KeyGen}$

$b \stackrel{\$}{\leftarrow} \{0,1\}$

// $(K_0^*, c^*) \leftarrow \text{Encaps}(pk)$:

$m^* \stackrel{\$}{\leftarrow} \mathcal{M}$

$c^* \leftarrow \text{Enc}(pk, m^*)$

$K_0^* \stackrel{\$}{\leftarrow} \mathcal{K}$

$T_H := T_H \cup \{(m^*, c^*, K_0^*)\}$

$K_1^* \stackrel{\$}{\leftarrow} \mathcal{K}$

return pk, c^*, K_b^*

$H(m, c)$:

if $(m, c) = (m^*, c^*)$
and query was by $\mathcal{A}$

CHALL := true

Abort

if $\exists K$ such that
 $(m, c, K) \in T_H$

return K

$K \stackrel{\$}{\leftarrow} \mathcal{K}$

$T_H := T_H \cup \{(m, c, K)\}$

return K

Finalize ($b' \in \{0,1\}$):

return $b' =_? b$

Fujisaki-Okamoto

Step 1: U-transform - OW PKE to IND-CPA KEM

We now show: Since b is independent of $\mathcal{A}$'s view,

$$\Pr[G_1^{\mathcal{A}} \Rightarrow 1] = 1/2 .$$

We rewrite G_1 one more time: in Initialize, we no longer store $K_0^* \xleftarrow{\$} \mathcal{K}$ in T_H for m^*, c^* .

The success probabilities do not change:

This entry in T_H is exclusively used in H , and only if H is queried on (m^*, c^*) .

$\mathcal{A}$ cannot succeed if the game aborts due to CHALL.

So if $\mathcal{A}$ succeeds, then this entry in T_H will not have been used anywhere.

We notice: with this rewrite, there is no difference left between $K_0^* \xleftarrow{\$} \mathcal{K}$ and $K_1^* \xleftarrow{\$} \mathcal{K}$.

Games G_0 and G_1

Initialize:

$(pk, sk) \leftarrow \text{KeyGen}$

$b \xleftarrow{\$} \{0,1\}$

// $(K_0^*, c^*) \leftarrow \text{Encaps}(pk)$:

$m^* \xleftarrow{\$} \mathcal{M}$

$c^* \leftarrow \text{Enc}(pk, m^*)$

$K_0^* \xleftarrow{\$} \mathcal{K}$

$T_H := T_H \cup \{(m^*, c^*, K_0^*)\}$

$K_1^* \xleftarrow{\$} \mathcal{K}$

return pk, c^*, K_b^*

$H(m, c)$:

if $(m, c) = (m^*, c^*)$
and query was by $\mathcal{A}$

CHALL := true

Abort

if $\exists K$ such that
 $(m, c, K) \in T_H$

return K

$K \xleftarrow{\$} \mathcal{K}$

$T_H := T_H \cup \{(m, c, K)\}$

return K

Finalize ($b' \in \{0,1\}$):

return $b' =_? b$

Fujisaki-Okamoto

Step 1: U-transform - OW PKE to IND-CPA KEM

We now show: Since b is independent of $\mathcal{A}$'s view,

$$\Pr[G_1^{\mathcal{A}} \Rightarrow 1] = 1/2 .$$

We rewrite G_1 one last time: in Initialize, we no longer sample K_0^* and K_1^* and then return K_b^* .

Instead, we just sample a single K^* and return that.

The success probabilities do not change:

It makes no difference if we flip a bit and then pick one of two uniform keys, or if we just pick a uniform key.

We notice: with this rewrite, b is no longer used anywhere, so “independent of $\mathcal{A}$'s view”.

Games G_0 and G_1

Initialize:

$(pk, sk) \leftarrow \text{KeyGen}$

$b \xleftarrow{\$} \{0,1\}$

$//(K_0^*, c^*) \leftarrow \text{Encaps}(pk):$

$m^* \xleftarrow{\$} \mathcal{M}$

$c^* \leftarrow \text{Enc}(pk, m^*)$

~~$K_0^* \xleftarrow{\$} \mathcal{K}$~~

~~$K_1^* \xleftarrow{\$} \mathcal{K}$~~

$K^* \xleftarrow{\$} \mathcal{K}$

return $pk, c^*, \cancel{K_b^*} K^*$

$H(m, c):$

if $(m, c) = (m^*, c^*)$
and query was by $\mathcal{A}$

CHALL := true

Abort

if $\exists K$ such that
 $(m, c, K) \in T_H$

return K

$K \xleftarrow{\$} \mathcal{K}$

$T_H := T_H \cup \{(m, c, K)\}$

return K

Finalize ($b' \in \{0,1\}$):

return $b' =_? b$

Fujisaki-Okamoto

Step 1: U-transform - OW PKE to IND-CPA KEM

We now show: Since b is independent of $\mathcal{A}$'s view,

$$\Pr[G_1^{\mathcal{A}} \Rightarrow 1] = 1/2 .$$

Let $b' = b$ denote the event “ $\mathcal{A}$ wins in G_1 ”.

We condition on whatever value b' is put out by $\mathcal{A}$:

$$\Pr[b' = b] = \sum_{\beta \in \{0,1\}} \Pr[\beta = b \mid b' = \beta] \cdot \Pr[b' = \beta] .$$

Since b and b' are independent, for both options β

$$\Pr[b = \beta \mid b' = \beta] = \Pr[b = \beta] = \frac{1}{2} .$$

Therefore

$$\Pr[b' = b] = \sum_{\beta \in \{0,1\}} \frac{1}{2} \cdot \Pr[b' = \beta] = \frac{1}{2} \sum_{\beta \in \{0,1\}} \Pr[b' = \beta] ,$$

And $\sum_{\beta \in \{0,1\}} \Pr[b' = \beta] = 1$ no matter how $\mathcal{A}$ looks.

Games G_0 and G_1

Initialize:

$(pk, sk) \leftarrow \text{KeyGen}$

$b \xleftarrow{\$} \{0,1\}$

$// (K_0^*, c^*) \leftarrow \text{Encaps}(pk):$

$m^* \xleftarrow{\$} \mathcal{M}$

$c^* \leftarrow \text{Enc}(pk, m^*)$

$K^* \xleftarrow{\$} \mathcal{K}$

return pk, c^*, K^*

Finalize ($b' \in \{0,1\}$):

return $b' =_? b$

$H(m, c):$

if $(m, c) = (m^*, c^*)$
and query was by $\mathcal{A}$

CHALL := true

Abort

if $\exists K$ such that
 $(m, c, K) \in T_H$

return K

$K \xleftarrow{\$} \mathcal{K}$

$T_H := T_H \cup \{(m, c, K)\}$

return K

Fujisaki-Okamoto

Step 1: U-transform - OW PKE to IND-CPA KEM

Security claim (PKE OW $\Rightarrow$ KEM[⊥] IND-CPA)

For any IND-CPA adversary $\mathcal{A}$ against KEM[⊥], issuing at most q_H queries to random oracle H , there exists an OW adversary $\mathcal{B}$ against PKE such that

$$\text{Adv}_{\text{KEM}^\perp}^{\text{IND-CPA}}(\mathcal{A}) \leq q_H \cdot \text{Adv}_{\text{PKE}}^{\text{OW}}(\mathcal{B}).$$

$\mathcal{B}$ runs in about the same time as $\mathcal{A}$.

Game analysis:

G_1 : $K_0 = H(m^*, c^*)$ is not given to $\mathcal{A}$ by the RO, so bit b is independent from $\mathcal{A}$'s view. We just saw:

$$\Pr[G_1^{\mathcal{A}} \Rightarrow 1] = 1/2,$$

so

$$\text{Adv}_{\text{KEM}}^{\text{IND-CPA}}(\mathcal{A}) \leq \Pr[\text{CHALL}].$$

Games G_0 and G_1

Initialize:

$(pk, sk) \leftarrow \text{KeyGen}$

$b \xleftarrow{\$} \{0,1\}$

// $(K_0^*, c^*) \leftarrow \text{Encaps}(pk)$:

$m^* \xleftarrow{\$} \mathcal{M}$

$c^* \leftarrow \text{Enc}(pk, m^*)$

$K_0^* := H(m^*, c^*)$

$K_1^* \xleftarrow{\$} \mathcal{K}$

return pk, c^*, K_b^*

Finalize ($b' \in \{0,1\}$):

return $b' =_? b$

$H(m, c)$:

if $(m, c) = (m^*, c^*)$
and query was by $\mathcal{A}$

CHALL := true

Abort

if $\exists K$ such that

$(m, c, K) \in T_H$

return K

$K \xleftarrow{\$} \mathcal{K}$

$T_H := T_H \cup \{(m, c, K)\}$

return K

Fujisaki-Okamoto

Step 1: U-transform - OW PKE to IND-CPA KEM

Security claim (PKE OW $\Rightarrow$ KEM[⊥] IND-CPA)

For any IND-CPA adversary $\mathcal{A}$ against KEM[⊥], issuing at most q_H queries to random oracle H , there exists an OW adversary $\mathcal{B}$ against PKE such that

$$\text{Adv}_{\text{KEM}^\perp}^{\text{IND-CPA}}(\mathcal{A}) \leq q_H \cdot \text{Adv}_{\text{PKE}}^{\text{OW}}(\mathcal{B}).$$

$\mathcal{B}$ runs in about the same time as $\mathcal{A}$.

Game analysis:

We bound $\Pr[\text{CHALL}]$ via OW reduction $\mathcal{B}$:

$\mathcal{B}$ simulates G_0/G_1 perfectly until CHALL happens.

If CHALL happens, then $(m^*, c^*, K) \in T_H$ for some K .

$\mathcal{B}$ terminates as follows: pick $(m', c^*, K') \xleftarrow{\$} T_H$, call Finalize(of the OW game) on m' .

How $\mathcal{B}$ simulates G_0

Initialize:

$pk, c^* \leftarrow \text{Init of OW}_{\text{PKE}}$

$b \xleftarrow{\$} \{0,1\}$

$H(K_0^*, c^*) \leftarrow \text{Encaps}(pk)$:

$m^* \xleftarrow{\$} \mathcal{M}$

$c^* \leftarrow \text{Enc}(pk, m^*)$

$K_0^* := H(m^*, c^*)$

$K^* \xleftarrow{\$} \mathcal{K}$

return pk, c^*, K^*

Finalize ($b' \in \{0,1\}$):

return $b' =_? b$

$H(m, c)$:

if $(m, c) = (m^*, c^*)$

and query was by $\mathcal{A}$

—CHALL := true

—Abort

if $\exists K$ such that

$(m, c, K) \in T_H$

return K

$K \xleftarrow{\$} \mathcal{K}$

$T_H := T_H \cup \{(m, c, K)\}$

return K

Fujisaki-Okamoto

Step 1: U-transform - OW PKE to IND-CPA KEM

Security claim (PKE OW $\Rightarrow$ KEM[⊥] IND-CPA)

For any IND-CPA adversary $\mathcal{A}$ against KEM[⊥], issuing at most q_H queries to random oracle H , there exists an OW adversary $\mathcal{B}$ against PKE such that

$$\text{Adv}_{\text{KEM}^\perp}^{\text{IND-CPA}}(\mathcal{A}) \leq q_H \cdot \text{Adv}_{\text{PKE}}^{\text{OW}}(\mathcal{B}).$$

$\mathcal{B}$ runs in about the same time as $\mathcal{A}$.

Game analysis:

We bound $\Pr[\text{CHALL}]$ via OW reduction $\mathcal{B}$:

$\mathcal{B}$ terminates with m' , where $(m', c^*, K') \xleftarrow{\$} T_H$.

If **CHALL** happens, $\mathcal{B}$ returns the correct $m' = m^*$ with probability $1/q_H$.

$$\Pr[\text{CHALL}] \leq q_H \cdot \text{Adv}_{\text{PKE}}^{\text{OW}}(\mathcal{B}).$$

How $\mathcal{B}$ simulates G_0

Initialize:

$pk, c^* \leftarrow \text{Init of OW}_{\text{PKE}}$

~~$b \xleftarrow{\$} \{0,1\}$~~

~~$H(K_0^*, c^*) \leftarrow \text{Encaps}(pk):$~~

~~$m^* \xleftarrow{\$} \mathcal{M}$~~

~~$c^* \leftarrow \text{Enc}(pk, m^*)$~~

~~$K_0^* := H(m^*, c^*)$~~

$K^* \xleftarrow{\$} \mathcal{K}$

return pk, c^*, K^*

Finalize ($b' \in \{0,1\}$):

return $b' =_? b$

$H(m, c)$:

~~if $(m, c) = (m^*, c^*)$~~

~~and query was by $\mathcal{A}$~~

~~—CHALL := true~~

~~—Abort~~

if $\exists K$ such that

$(m, c, K) \in T_H$

return K

$K \xleftarrow{\$} \mathcal{K}$

$T_H := T_H \cup \{(m, c, K)\}$

return K

Fujisaki-Okamoto

Step 1: U-transform - OW PKE to IND-CPA KEM

Security claim (PKE OW $\Rightarrow$ KEM[⊥] IND-CPA)

For any IND-CPA adversary $\mathcal{A}$ against KEM[⊥], issuing at most q_H queries to random oracle H , there exists an OW adversary $\mathcal{B}$ against PKE such that

$$\text{Adv}_{\text{KEM}^\perp}^{\text{IND-CPA}}(\mathcal{A}) \leq q_H \cdot \text{Adv}_{\text{PKE}}^{\text{OW}}(\mathcal{B}).$$

$\mathcal{B}$ runs in about the same time as $\mathcal{A}$.

Key insights: We used:

- **Lazy sampling** to simulate H .
- **Unpredictability:** unless $\mathcal{A}$ queries H on (m^*, c^*) , $K_0^* := H(m^*, c^*)$ could not possibly be distinguished from K_1^* .
- **“Observable” ROM:** $\mathcal{B}$ can observe RO queries and find (m^*, c^*) , the OW solution

How $\mathcal{B}$ simulates G_0

Initialize:

$pk, c^* \leftarrow \text{Init of OW}_{\text{PKE}}$

$b \xleftarrow{\$} \{0,1\}$

$H(K_0^*, c^*) \leftarrow \text{Encaps}(pk):$

$m^* \xleftarrow{\$} \mathcal{M}$

$c^* \leftarrow \text{Enc}(pk, m^*)$

$K_0^* := H(m^*, c^*)$

$K^* \xleftarrow{\$} \mathcal{K}$

return pk, c^*, K^*

Finalize ($b' \in \{0,1\}$):

return $b' =_? b$

$H(m, c):$

if $(m, c) = (m^*, c^*)$

and query was by $\mathcal{A}$

—CHALL := true

—Abort

if $\exists K$ such that

$(m, c, K) \in T_H$

return K

$K \xleftarrow{\$} \mathcal{K}$

$T_H := T_H \cup \{(m, c, K)\}$

return K

What is a KEM?

“Active” security definition for KEMs: IND-CCA

IND-CCA captures “KEM output keys look random, even under chosen-ciphertext attacks”. $\mathcal{A}$ also gets a decapsulation oracle.

Definition IND-CCA security

We define the **IND-CCA advantage** of an adversary $\mathcal{A}$ against KEM as

$$\text{Adv}_{\text{KEM}}^{\text{IND-CCA}}(\mathcal{A}) := \left| \Pr[\text{IND-CCA}^{\mathcal{A}} \Rightarrow 1] - \frac{1}{2} \right|.$$

Game $\text{IND-CCA}_{\text{KEM}}^{\mathcal{A}}$

Initialize:

$(pk, sk) \leftarrow \text{KeyGen}$

$b \xleftarrow{\$} \{0,1\}$

$(K_0^*, c^*) \leftarrow \text{Encaps}(pk)$

$K_1^* \xleftarrow{\$} \mathcal{K}$

return pk, c^*, K_b^*

Decaps($c \neq c^*$):

return $K := \text{Decaps}(sk, c)$

Finalize ($b' \in \{0,1\}$):

return $b' =_? b$

Fujisaki-Okamoto

Step 2: U-transform - PKE to IND-C_A KEM

Security claim (PKE OW $\Rightarrow$ KEM[⊥] IND-C_A)

For any IND-C_A adversary $\mathcal{A}$ against KEM[⊥], issuing at most q_H queries to random oracle H , there exists an OW adversary $\mathcal{B}$ against PKE such that

$$\text{Adv}_{\text{KEM}^\perp}^{\text{IND-C}_A}(\mathcal{A}) \leq q_H \cdot \text{Adv}_{\text{PKE}}^{\text{OW}}(\mathcal{B}).$$

$\mathcal{B}$ runs in about the same time as $\mathcal{A}$.

Main idea:

Only difference to IND-CPA proof: $\mathcal{A}$ now also has oracle **Decaps**.

How do we adapt the proof? We change **Decaps** so that reduction $\mathcal{B}$ can simulate it to $\mathcal{A}$.

But $\mathcal{B}$ doesn't have sk . What can we do? Lazy sampling?

Game IND-C_A ^{$\mathcal{A}$} _{KEM[⊥]}

Initialize:

$(pk, sk) \leftarrow \text{KeyGen}$

$b \xleftarrow{\$} \{0,1\}$

// $(K_0^*, c^*) \leftarrow \text{Encaps}(pk)$:

$m^* \xleftarrow{\$} \mathcal{M}$

$c^* \leftarrow \text{Enc}(pk, m^*)$

$K_0^* := H(m^*, c^*)$

$K_1^* \xleftarrow{\$} \mathcal{K}$

return pk, c^*, K_b^*

Finalize ($b' \in \{0,1\}$):

return $b' =_? b$

$H(m, c)$:

...

if $\exists K$ such that
 $(m, c, K) \in T_H$

return K

$K \xleftarrow{\$} \mathcal{K}$

$T_H := T_H \cup \{(m, c, K)\}$

Decaps($c \neq c^*$):

// $K := \text{Decaps}(sk, c)$:

$m' := \text{Dec}(sk, c)$

if $m' = \perp$ return $\perp$

return $K := H(m, c)$

Fujisaki-Okamoto

Step 2: U-transform - PKE to IND-C_A KEM

Security claim (PKE OW $\Rightarrow$ KEM[⊥] IND-C_A)

For any IND-C_A adversary $\mathcal{A}$ against KEM[⊥], issuing at most q_H queries to random oracle H , there exists an OW adversary $\mathcal{B}$ against PKE such that

$$\text{Adv}_{\text{KEM}^\perp}^{\text{IND-C}_A}(\mathcal{A}) \leq q_H \cdot \text{Adv}_{\text{PKE}}^{\text{OW}}(\mathcal{B}).$$

$\mathcal{B}$ runs in about the same time as $\mathcal{A}$.

Main idea:

Only difference to IND-CPA proof: $\mathcal{A}$ now also has oracle **Decaps**.

How do we adapt the proof? We change **Decaps** so that reduction $\mathcal{B}$ can simulate it to $\mathcal{A}$.

But $\mathcal{B}$ doesn't have sk . What can we do? Lazy sampling?

Game IND-C_A ^{$\mathcal{A}$} _{KEM[⊥]}

Initialize:

...

Finalize ($b' \in \{0,1\}$):

...

$H(m, c)$:

...

Decaps($c \neq c^*$):

$m' := \text{Dec}(sk, c)$

if $m' = \perp$ return $\perp$

return $K := H(m, c)$

Attempt 1: Simulating **Decaps**

if $\exists K$ such that $(c, K) \in T_D$

return K

$K \stackrel{\$}{\leftarrow} \mathcal{K}$

$T_D := T_D \cup \{(c, K)\}$

return K

Fujisaki-Okamoto

Step 2: U-transform - PKE to IND-C_A KEM

Security claim (PKE OW $\Rightarrow$ KEM[⊥] IND-C_A)

For any IND-C_A adversary $\mathcal{A}$ against KEM[⊥], issuing at most q_H queries to random oracle H , there exists an OW adversary $\mathcal{B}$ against PKE such that

$$\text{Adv}_{\text{KEM}^\perp}^{\text{IND-C}_A}(\mathcal{A}) \leq q_H \cdot \text{Adv}_{\text{PKE}}^{\text{OW}}(\mathcal{B}).$$

$\mathcal{B}$ runs in about the same time as $\mathcal{A}$.

Game IND-C_A ^{$\mathcal{A}$} _{KEM[⊥]}

Initialize:

...

$H(m, c)$:

...

Finalize ($b' \in \{0,1\}$):

...

Decaps($c \neq c^*$):

$m' := \text{Dec}(\text{sk}, c)$

if $m' = \perp$ return $\perp$

return $K := H(m, c)$

Question

Is this simulation convincing?

Attempt 1: Simulating Decaps

if $\exists K$ such that $(c, K) \in T_D$

return K

$K \stackrel{\$}{\leftarrow} \mathcal{K}$

$T_D := T_D \cup \{(c, K)\}$

return K

Fujisaki-Okamoto

Step 2: U-transform - PKE to IND-C_A KEM

Security claim (PKE OW $\Rightarrow$ KEM[⊥] IND-C_A)

For any IND-C_A adversary $\mathcal{A}$ against KEM[⊥], issuing at most q_H queries to random oracle H , there exists an OW adversary $\mathcal{B}$ against PKE such that

$$\text{Adv}_{\text{KEM}^\perp}^{\text{IND-C}_A}(\mathcal{A}) \leq q_H \cdot \text{Adv}_{\text{PKE}}^{\text{OW}}(\mathcal{B}).$$

$\mathcal{B}$ runs in about the same time as $\mathcal{A}$.

Simulation flaw: “Everything is so random here.”

Decaps is not independent of H ! In KEM[⊥]:

$$\text{Decaps}(\text{sk}, c) = H(\text{Dec}(\text{sk}, c), c)$$

and we fully break this connection!

$\mathcal{A}$ immediately notices our lazily-sampled simulation when querying $H(m, c)$ for any $m \in \mathcal{M}$ and $c \leftarrow \text{Enc}(\text{pk}, m)$, and then Decaps on c .

Game IND-C_A ^{$\mathcal{A}$} _{KEM[⊥]}

Initialize:

...

Finalize ($b' \in \{0,1\}$):

...

$H(m, c)$:

...

Decaps($c \neq c^*$):

$m' := \text{Dec}(\text{sk}, c)$

if $m' = \perp$ return $\perp$

return $K := H(m, c)$

Attempt 1: Simulating Decaps

if $\exists K$ such that $(c, K) \in T_D$

return K

$K \stackrel{\$}{\leftarrow} \mathcal{K}$

$T_D := T_D \cup \{(c, K)\}$

return K

Fujisaki-Okamoto

Step 2: U-transform - PKE to IND-C_A KEM

Security claim (PKE OW $\Rightarrow$ KEM[⊥] IND-C_A)

For any IND-C_A adversary $\mathcal{A}$ against KEM[⊥], issuing at most q_H queries to random oracle H , there exists an OW adversary $\mathcal{B}$ against PKE such that

$$\text{Adv}_{\text{KEM}^\perp}^{\text{IND-C}_A}(\mathcal{A}) \leq q_H \cdot \text{Adv}_{\text{PKE}}^{\text{OW}}(\mathcal{B}).$$

$\mathcal{B}$ runs in about the same time as $\mathcal{A}$.

Simulation flaw: “Everything is so random here.”

What is the problem? That neither **Decaps** nor H notice when they should use the same K .

What if H would

- notice when c decrypts to m , and
- ensure consistency with **Decaps** via an additional table T_D ?

Game IND-C_A ^{$\mathcal{A}$} _{KEM[⊥]}

Initialize:

...

Finalize ($b' \in \{0,1\}$):

...

$H(m, c)$:

...

Decaps($c \neq c^*$):

$m' := \text{Dec}(\text{sk}, c)$

if $m' = \perp$ return $\perp$

return $K := H(m, c)$

Attempt 1: Simulating **Decaps**

if $\exists K$ such that $(c, K) \in T_D$

return K

$K \xleftarrow{\$} \mathcal{K}$

$T_D := T_D \cup \{(c, K)\}$

return K

Fujisaki-Okamoto

Step 2: U-transform - PKE to IND-CCA KEM

Security claim (PKE OW $\Rightarrow$ KEM[⊥] IND-CCA)

For any IND-CCA adversary $\mathcal{A}$ against KEM[⊥], issuing at most q_H queries to random oracle H , there exists an OW adversary $\mathcal{B}$ against PKE such that

$$\text{Adv}_{\text{KEM}^\perp}^{\text{IND-CCA}}(\mathcal{A}) \leq q_H \cdot \text{Adv}_{\text{PKE}}^{\text{OW}}(\mathcal{B}).$$

$\mathcal{B}$ runs in about the same time as $\mathcal{A}$.

Simulation flaw: “Everything is so random here.”

What is the problem? That neither Decaps nor H notice when they should use the same K .

What if H would

- notice when c decrypts to m , and
- ensure consistency with Decaps via an additional table T_D ?

Decaps and H in game IND-CCA ^{$\mathcal{A}$} _{KEM[⊥]}

<u>Decaps</u> ($c \neq c^*$):	<u>$H(m, c)$</u> :
$m' := \text{Dec}(\text{sk}, c)$	if $\exists K$ such that $(m, c, K) \in T_H$
if $m' = \perp$	return K
return $\perp$	$K \xleftarrow{\$} \mathcal{K}$
$K := H(m, c)$	$T_H := T_H \cup \{(m, c, K)\}$
return K	return K

Attempt 2: Simulating H and Decaps

<u>Decaps</u> ($c \neq c^*$):	<u>$H(m, c)$</u> :
if $\exists K$ such	if $\exists K$ such that $(m, c, K) \in T_H$
that $(c, K) \in T_D$	return K
return K	$K \xleftarrow{\$} \mathcal{K}$
$K \xleftarrow{\$} \mathcal{K}$	if $\text{Dec}(\text{sk}, c) = m$
$T_D := T_D$	if $\exists K'$ such that $(c, K') \in T_D$
$\cup \{(c, K)\}$	$K := K'$
return K	else
	$T_D := T_D \cup \{(c, K)\}$
	$T_H := T_H \cup \{(m, c, K)\}$
	return K

Fujisaki-Okamoto

Step 2: U-transform - PKE to IND-CCA KEM

Security claim (PKE OW $\Rightarrow$ KEM[⊥] IND-CCA)

For any IND-CCA adversary $\mathcal{A}$ against KEM[⊥], issuing at most q_H queries to random oracle H , there exists an OW adversary $\mathcal{B}$ against PKE such that

$$\text{Adv}_{\text{KEM}^\perp}^{\text{IND-CCA}}(\mathcal{A}) \leq q_H \cdot \text{Adv}_{\text{PKE}}^{\text{OW}}(\mathcal{B}).$$

$\mathcal{B}$ runs in about the same time as $\mathcal{A}$.

Question

Is this simulation convincing?

Decaps and H in game IND-CCA ^{$\mathcal{A}$} _{KEM[⊥]}

<u>Decaps</u> ($c \neq c^*$):	<u>$H(m, c)$</u> :
$m' := \text{Dec}(\text{sk}, c)$	if $\exists K$ such that $(m, c, K) \in T_H$
if $m' = \perp$	return K
return $\perp$	$K \xleftarrow{\$} \mathcal{K}$
$K := H(m, c)$	$T_H := T_H \cup \{(m, c, K)\}$
return K	return K

Attempt 2: Simulating H and Decaps

<u>Decaps</u> ($c \neq c^*$):	<u>$H(m, c)$</u> :
if $\exists K$ such	if $\exists K$ such that $(m, c, K) \in T_H$
that $(c, K) \in T_D$	return K
return K	$K \xleftarrow{\$} \mathcal{K}$
$K \xleftarrow{\$} \mathcal{K}$	if $\text{Dec}(\text{sk}, c) = m$
$T_D := T_D$	if $\exists K'$ such that $(c, K') \in T_D$
$\cup \{(c, K)\}$	$K := K'$
return K	else
	$T_D := T_D \cup \{(c, K)\}$
	$T_H := T_H \cup \{(m, c, K)\}$
	return K

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Step 2: U-transform - PKE to IND-CCA KEM

Security claim (PKE OW $\Rightarrow$ KEM[⊥] IND-CCA)

For any IND-CCA adversary $\mathcal{A}$ against KEM[⊥], issuing at most q_H queries to random oracle H , there exists an OW adversary $\mathcal{B}$ against PKE such that

$$Adv_{KEM^\perp}^{IND-CCA}(\mathcal{A}) \leq q_H \cdot Adv_{PKE}^{OW}(\mathcal{B}).$$

$\mathcal{B}$ runs in about the same time as $\mathcal{A}$.

Simulation flaw: “Where did the bouncer go?”

In KEM[⊥], Decaps behaves like a “bouncer”: if c is invalid, then Decaps(c) = ⊥.

Our simulated Decaps does not do this correctly: every ciphertext gets a key.

Decaps and H in game IND-CCA ^{$\mathcal{A}$} _{KEM[⊥]}

<u>Decaps($c \neq c^*$):</u>	<u>$H(m, c)$:</u>
$m' := \text{Dec}(\text{sk}, c)$	if $\exists K$ such that $(m, c, K) \in T_H$
if $m' = \perp$	return K
return $\perp$	$K \xleftarrow{\$} \mathcal{K}$
$K := H(m, c)$	$T_H := T_H \cup \{(m, c, K)\}$
return K	return K

Attempt 2: Simulating H and Decaps

<u>Decaps($c \neq c^*$):</u>	<u>$H(m, c)$:</u>
if $\exists K$ such	if $\exists K$ such that $(m, c, K) \in T_H$
that $(c, K) \in T_D$	return K
return K	$K \xleftarrow{\$} \mathcal{K}$
$K \xleftarrow{\$} \mathcal{K}$	if Dec(sk, c) = m
$T_D := T_D$	if $\exists K'$ such that $(c, K') \in T_D$
$\cup \{(c, K)\}$	$K := K'$
return K	else
	$T_D := T_D \cup \{(c, K)\}$
	$T_H := T_H \cup \{(m, c, K)\}$
	return K

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Step 2: U-transform - PKE to IND-CCA KEM

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$$\text{Adv}_{\text{KEM}^\perp}^{\text{IND-CCA}}(\mathcal{A}) \leq q_H \cdot \text{Adv}_{\text{PKE}}^{\text{OW}}(\mathcal{B}).$$

$\mathcal{B}$ runs in about the same time as $\mathcal{A}$.

Simulation flaw: “Where did the bouncer go?”

What is the problem? That **Decaps** does not react correctly to invalid ciphertexts.

What if **Decaps** would

- notice such ciphertexts, and
- reject?

Decaps and H in game IND-CCA ^{$\mathcal{A}$} _{KEM[⊥]}

Decaps ($c \neq c^*$):	H (m, c):
$m' := \text{Dec}(\text{sk}, c)$	if $\exists K$ such that $(m, c, K) \in T_H$
if $m' = \perp$	return K
return $\perp$	$K \xleftarrow{\$} \mathcal{K}$
$K := H(m, c)$	$T_H := T_H \cup \{(m, c, K)\}$
return K	return K

Attempt 2: Simulating H and **Decaps**

Decaps ($c \neq c^*$):	H (m, c):
if $\exists K$ such	if $\exists K$ such that $(m, c, K) \in T_H$
that $(c, K) \in T_D$	return K
return K	$K \xleftarrow{\$} \mathcal{K}$
$K \xleftarrow{\$} \mathcal{K}$	if $\text{Dec}(\text{sk}, c) = m$
$T_D := T_D$	if $\exists K'$ such that $(c, K') \in T_D$
$\cup \{(c, K)\}$	$K := K'$
return K	else
	$T_D := T_D \cup \{(c, K)\}$
	$T_H := T_H \cup \{(m, c, K)\}$
	return K

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Step 2: U-transform - PKE to IND-CCA KEM

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$\mathcal{B}$ runs in about the same time as $\mathcal{A}$.

Simulation flaw: “Where did the bouncer go?”

What is the problem? That Decaps does not react correctly to invalid ciphertexts.

What if Decaps would

- notice such ciphertexts, and
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Decaps and H in game IND-CCA ^{$\mathcal{A}$} _{KEM[⊥]}

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if $m' = \perp$	return K
return $\perp$	$K \xleftarrow{\$} \mathcal{K}$
$K := H(m, c)$	$T_H := T_H \cup \{(m, c, K)\}$
return K	return K

Attempt 3: Simulating H and Decaps

<u>Decaps</u> ($c \neq c^*$):	<u>$H(m, c)$</u> :
if $\text{Dec}(\text{sk}, c) \notin \mathcal{M}$	if $\exists K$ such that $(m, c, K) \in T_H$
return $\perp$	return K
if $\exists K$ such that $(c, K) \in T_D$	$K \xleftarrow{\$} \mathcal{K}$
return K	if $\text{Dec}(\text{sk}, c) = m$
$K \xleftarrow{\$} \mathcal{K}$	if $\exists K'$ such that $(c, K') \in T_D$
$T_D := T_D \cup \{(c, K)\}$	$K := K'$
return K	else
	$T_D := T_D \cup \{(c, K)\}$
	$T_H := T_H \cup \{(m, c, K)\}$
	return K

Fujisaki-Okamoto

Step 2: U-transform - PKE to IND-CCA KEM

Security claim (PKE OW $\Rightarrow$ KEM[⊥] IND-CCA)

For any IND-CCA adversary $\mathcal{A}$ against KEM[⊥], issuing at most q_H queries to random oracle H , there exists an OW adversary $\mathcal{B}$ against PKE such that

$$\text{Adv}_{\text{KEM}^\perp}^{\text{IND-CCA}}(\mathcal{A}) \leq q_H \cdot \text{Adv}_{\text{PKE}}^{\text{OW}}(\mathcal{B}).$$

$\mathcal{B}$ runs in about the same time as $\mathcal{A}$.

Question

Is this simulation convincing?

Decaps and H in game IND-CCA ^{$\mathcal{A}$} _{KEM[⊥]}

<u>Decaps</u> ($c \neq c^*$):	<u>$H(m, c)$</u> :
$m' := \text{Dec}(\text{sk}, c)$	if $\exists K$ such that $(m, c, K) \in T_H$
if $m' = \perp$	return K
return $\perp$	$K \xleftarrow{\$} \{0,1\}^n$
$K := H(m, c)$	$T_H := T_H \cup \{(m, c, K)\}$
return K	return K

Attempt 3: Simulating H and Decaps

<u>Decaps</u> ($c \neq c^*$):	<u>$H(m, c)$</u> :
if $\text{Dec}(\text{sk}, c) \notin \mathcal{M}$	if $\exists K$ such that $(m, c, K) \in T_H$
return $\perp$	return K
if $\exists K$ such that $(c, K) \in T_D$	$K \xleftarrow{\$} \mathcal{K}$
return K	if $\text{Dec}(\text{sk}, c) = m$
$K \xleftarrow{\$} \mathcal{K}$	if $\exists K'$ such that $(c, K') \in T_D$
$T_D := T_D \cup \{(c, K)\}$	$K := K'$
return K	else
	$T_D := T_D \cup \{(c, K)\}$
	$T_H := T_H \cup \{(m, c, K)\}$
	return K

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Step 2: U-transform - PKE to IND-CCA KEM

Security claim (PKE OW $\Rightarrow$ KEM[⊥] IND-CCA)

For any IND-CCA adversary $\mathcal{A}$ against KEM[⊥], issuing at most q_H queries to random oracle H , there exists an OW adversary $\mathcal{B}$ against PKE such that

$$\text{Adv}_{\text{KEM}^\perp}^{\text{IND-CCA}}(\mathcal{A}) \leq q_H \cdot \text{Adv}_{\text{PKE}}^{\text{OW}}(\mathcal{B}).$$

$\mathcal{B}$ runs in about the same time as $\mathcal{A}$.

Question

Is this simulation convincing?

Actually yes, but: How could $\mathcal{B}$ simulate this?

$\mathcal{B}$ does not have sk .

Decaps and H in game IND-CCA ^{$\mathcal{A}$} _{KEM[⊥]}

<u>Decaps</u> ($c \neq c^*$):	<u>$H(m, c)$</u> :
$m' := \text{Dec}(sk, c)$	if $\exists K$ such that $(m, c, K) \in T_H$
if $m' = \perp$	return K
return $\perp$	$K \xleftarrow{\$} \{0,1\}^n$
$K := H(m, c)$	$T_H := T_H \cup \{(m, c, K)\}$
return K	return K

Attempt 3: Simulating H and Decaps

<u>Decaps</u> ($c \neq c^*$):	<u>$H(m, c)$</u> :
if $\text{Dec}(sk, c) \notin \mathcal{M}$	if $\exists K$ such that $(m, c, K) \in T_H$
return $\perp$	return K
if $\exists K$ such that $(c, K) \in T_D$	$K \xleftarrow{\$} \mathcal{K}$
return K	if $\text{Dec}(sk, c) = m$
$K \xleftarrow{\$} \mathcal{K}$	if $\exists K'$ such that $(c, K') \in T_D$
$T_D := T_D \cup \{(c, K)\}$	$K := K'$
return K	else
	$T_D := T_D \cup \{(c, K)\}$
	$T_H := T_H \cup \{(m, c, K)\}$
	return K

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Step 2: U-transform - PKE to IND-C_A KEM

Security claim (PKE OW $\Rightarrow$ KEM[⊥] IND-C_A)

For any IND-C_A adversary $\mathcal{A}$ against KEM[⊥], issuing at most q_H queries to random oracle H , there exists an OW adversary $\mathcal{B}$ against PKE such that

$$\text{Adv}_{\text{KEM}^\perp}^{\text{IND-C}_A}(\mathcal{A}) \leq q_H \cdot \text{Adv}_{\text{PKE}}^{\text{OW}}(\mathcal{B}).$$

$\mathcal{B}$ runs in about the same time as $\mathcal{A}$.

Question

Is this simulation convincing?

Actually yes, but: How could $\mathcal{B}$ simulate this?

$\mathcal{B}$ does not have sk .

Simulated H and Decaps

<u>Decaps</u> ($c \neq c^*$):	<u>$H(m, c)$</u> :
if $\text{Dec}(sk, c) \notin \mathcal{M}$	if $\exists K$ such that $(m, c, K) \in T_H$
return $\perp$	return K
if $\exists K$ such	$K \xleftarrow{\$} \mathcal{K}$
that $(c, K) \in T_D$	if $\text{Dec}(sk, c) = m$
return K	if $\exists K'$ such that $(c, K') \in T_D$
$K \xleftarrow{\$} \mathcal{K}$	$K := K'$
$T_D := T_D$	else
$\cup \{(c, K)\}$	$T_D := T_D \cup \{(c, K)\}$
return K	$T_H := T_H \cup \{(m, c, K)\}$
	return K

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Step 2: U-transform - PKE to IND-C_A KEM

Security claim (PKE OW $\Rightarrow$ KEM[⊥] IND-C_A)

For any IND-C_A adversary $\mathcal{A}$ against KEM[⊥], issuing at most q_H queries to random oracle H , there exists an OW adversary $\mathcal{B}$ against PKE such that

$$\text{Adv}_{\text{KEM}^\perp}^{\text{IND-C}_A}(\mathcal{A}) \leq q_H \cdot \text{Adv}_{\text{PKE}}^{\text{OW}}(\mathcal{B}).$$

$\mathcal{B}$ runs in about the same time as $\mathcal{A}$.

How could $\mathcal{B}$ simulate this without sk ?

What if $\mathcal{B}$ had two additional oracles?

Valid(c)

//check: is c valid?

$m' := \text{Dec}(sk, c)$

If $m' \in \mathcal{M}$ return true

else return false

PTcheck(m, c)

//check: does m fit c ?

$m' := \text{Dec}(sk, c)$

return $m' =_? m$

Simulated H and Decaps

<u>Decaps($c \neq c^*$):</u>	<u>$H(m, c)$:</u>
if $\text{Dec}(sk, c) \notin \mathcal{M}$	if $\exists K$ such that $(m, c, K) \in T_H$
return $\perp$	return K
if $\exists K$ such	$K \xleftarrow{\$} \mathcal{K}$
that $(c, K) \in T_D$	if $\text{Dec}(sk, c) = m$
return K	if $\exists K'$ such that $(c, K') \in T_D$
$K \xleftarrow{\$} \mathcal{K}$	$K := K'$
$T_D := T_D$	else
$\cup \{(c, K)\}$	$T_D := T_D \cup \{(c, K)\}$
return K	$T_H := T_H \cup \{(m, c, K)\}$
	return K

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Step 2: U-transform - PKE to IND-C_A KEM

Security claim (PKE OW $\Rightarrow$ KEM[⊥] IND-C_A)

For any IND-C_A adversary $\mathcal{A}$ against KEM[⊥], issuing at most q_H queries to random oracle H , there exists an OW adversary $\mathcal{B}$ against PKE such that

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$\mathcal{B}$ runs in about the same time as $\mathcal{A}$.

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//check: does m fit c ?

$m' := \text{Dec}(sk, c)$

return $m' =_? m$

Simulated H and Decaps

<u>Decaps($c \neq c^*$)</u> :	<u>$H(m, c)$</u> :
if not Valid(c)	if $\exists K$ such that $(m, c, K) \in T_H$
return $\perp$	return K
if $\exists K$ such	$K \xleftarrow{\$} \{0,1\}^n$
that $(c, K) \in T_D$	if PTcheck(m, c)
return K	if $\exists K'$ such that $(c, K') \in T_D$
$K \xleftarrow{\$} \mathcal{K}$	$K := K'$
$T_D := T_D$	else
$\cup \{(c, K)\}$	$T_D := T_D \cup \{(c, K)\}$
return K	$T_H := T_H \cup \{(m, c, K)\}$
	return K

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Step 2: U-transform - PKE to IND-C_A KEM

Security claim (PKE OW $\Rightarrow$ KEM[⊥] IND-C_A)

For any IND-C_A adversary $\mathcal{A}$ against KEM[⊥], issuing at most q_H queries to random oracle H , there exists an OW adversary $\mathcal{B}$ against PKE such that

$$Adv_{KEM^\perp}^{IND-C_A}(\mathcal{A}) \leq q_H \cdot Adv_{PKE}^{OW}(\mathcal{B}).$$

$\mathcal{B}$ runs in about the same time as $\mathcal{A}$.

How could $\mathcal{B}$ simulate this without sk ?

What if $\mathcal{B}$ had two additional oracles?

Valid(c)

//check: is c valid?

$m' := \text{Dec}(sk, c)$

If $m' \in \mathcal{M}$ return true

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PTcheck(m, c)

//check: does m fit c ?

$m' := \text{Dec}(sk, c)$

return $m' =_? m$

Simulated H and Decaps

<u>Decaps($c \neq c^*$)</u> :	<u>$H(m, c)$</u> :
if not Valid(c)	if $\exists K$ such that $(m, c, K) \in T_H$
return $\perp$	return K
if $\exists K$ such	$K \xleftarrow{\$} \{0,1\}^n$
that $(c, K) \in T_D$	if PTcheck(m, c)
return K	if $\exists K'$ such that $(c, K') \in T_D$
$K \xleftarrow{\$} \mathcal{K}$	$K := K'$
$T_D := T_D$	else
$\cup \{(c, K)\}$	$T_D := T_D \cup \{(c, K)\}$
return K	$T_H := T_H \cup \{(m, c, K)\}$
	return K

But: This needs adapting of our OW notion!

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Step 2: U-transform - PKE to IND-CCA KEM

We formalize the security property of PKE that we need for the proof:

One-Way security in the presence of oracles **PtCheck** and **Valid**.

Definition OW-PCVA security

We define the **OW-PCVA advantage** of an adversary $\mathcal{A}$ against PKE as

$$\text{Adv}_{\text{PKE}}^{\text{OW-PCVA}}(\mathcal{A}) := \Pr[\text{OWPCVA}_{\text{PKE}}^{\mathcal{A}} \Rightarrow 1].$$

Game $\text{OWPCVA}_{\text{PKE}}^{\mathcal{A}}$

Initialize:

$(pk, sk) \leftarrow \text{KeyGen}$

$m^* \xleftarrow{\$} \mathcal{M}$

$c^* \leftarrow \text{Enc}(pk, m^*)$

return pk, c^*

Finalize ($m' \in \mathcal{M}$):

return $m' =_? m^*$

Ptcheck(m, c)

$m' := \text{Dec}(sk, c)$

return $m' =_? m$

Valid(c)

$m' := \text{Dec}(sk, c)$

If $m' \in \mathcal{M}$ return true

else return false

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Step 2: U-transform - PKE to IND-CCA KEM

We correct the PKE requirement in our security claim:

Security claim (PKE **OW-PCVA** $\Rightarrow$ KEM[⊥] IND-CCA)

*For any IND-CCA adversary $\mathcal{A}$ against KEM[⊥], issuing at most q_H queries to random oracle H , there exists an **OW-PCVA** adversary $\mathcal{B}$ against PKE such that*

$$Adv_{KEM^\perp}^{IND-CCA}(\mathcal{A}) \leq q_H \cdot Adv_{PKE}^{OW-PCVA}(\mathcal{B}).$$

$\mathcal{B}$ runs in about the same time as $\mathcal{A}$.

The formal proof goes like this:

G_0 = IND-CPA game for KEM[⊥].

G_1 : Replace **Decaps**, H by the simulations 2 slides ago.

G_2 : Abort if **CHALL** happens, so if H gets queried on m^*, c^* .

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Step 2: U-transform - PKE to IND-CCA KEM

We correct the PKE requirement in our security claim:

Security claim (PKE OW-PCVA $\Rightarrow$ KEM $^\perp$ IND-CCA)

For any IND-CCA adversary $\mathcal{A}$ against KEM $^\perp$, issuing at most q_H queries to random oracle H , there exists an OW-PCVA adversary $\mathcal{B}$ against PKE such that

$$\text{Adv}_{\text{KEM}^\perp}^{\text{IND-CCA}}(\mathcal{A}) \leq q_H \cdot \text{Adv}_{\text{PKE}}^{\text{OW-PCVA}}(\mathcal{B}).$$

$\mathcal{B}$ runs in about the same time as $\mathcal{A}$.

Analysis:

G_0 = IND-CCA game for KEM $^\perp$.

$$\text{Adv}_{\text{KEM}^\perp}^{\text{IND-CCA}}(\mathcal{A}) = \left| \Pr[G_0^{\mathcal{A}} \Rightarrow 1] - \frac{1}{2} \right|.$$

G_1 : Replace Decaps, H by the simulations 2 slides ago.

$$\Pr[G_0^{\mathcal{A}} \Rightarrow 1] = \Pr[G_1^{\mathcal{A}} \Rightarrow 1].$$

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Step 2: U-transform - PKE to IND-C_A KEM

We correct the PKE requirement in our security claim:

Security claim (PKE OW-PCVA $\Rightarrow$ KEM[⊥] IND-C_A)

For any IND-C_A adversary $\mathcal{A}$ against KEM[⊥], issuing at most q_H queries to random oracle H , there exists an OW-PCVA adversary $\mathcal{B}$ against PKE such that

$$\text{Adv}_{\text{KEM}^\perp}^{\text{IND-C}_A}(\mathcal{A}) \leq q_H \cdot \text{Adv}_{\text{PKE}}^{\text{OW-PCVA}}(\mathcal{B}).$$

$\mathcal{B}$ runs in about the same time as $\mathcal{A}$.

Analysis:

G_1 : Replace Decaps, H by the simulations 2 slides ago.

G_2 : Abort if CHALL happens.

$$| \Pr[G_1^{\mathcal{A}} \Rightarrow 1] - \Pr[G_2^{\mathcal{A}} \Rightarrow 1] | \leq \Pr[\text{CHALL}] ,$$

so

$$\text{Adv}_{\text{KEM}^\perp}^{\text{IND-CCA}}(\mathcal{A}) \leq \Pr[\text{CHALL}] .$$

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Step 2: U-transform - PKE to IND-C_A KEM

We correct the PKE requirement in our security claim:

Security claim (PKE OW-PCVA $\Rightarrow$ KEM[⊥] IND-C_A)

For any IND-C_A adversary $\mathcal{A}$ against KEM[⊥], issuing at most q_H queries to random oracle H , there exists an OW-PCVA adversary $\mathcal{B}$ against PKE such that

$$Adv_{KEM^\perp}^{IND-C_A}(\mathcal{A}) \leq q_H \cdot Adv_{PKE}^{OW-PCVA}(\mathcal{B}).$$

$\mathcal{B}$ runs in about the same time as $\mathcal{A}$.

Analysis:

We bound $\Pr[\text{CHALL}]$ via the OW-PCVA adversary $\mathcal{B}$.

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Step 2: U-transform - PKE to IND-C_A KEM

We correct the PKE requirement in our security claim:

Security claim (PKE OW-PCVA $\Rightarrow$ KEM[⊥] IND-C_A)

For any IND-C_A adversary $\mathcal{A}$ against KEM[⊥], issuing at most q_H queries to random oracle H , there exists an OW-PCVA adversary $\mathcal{B}$ against PKE such that

$$\text{Adv}_{\text{KEM}^\perp}^{\text{IND-C}_A}(\mathcal{A}) \leq \text{Adv}_{\text{PKE}}^{\text{OW-PCVA}}(\mathcal{B}).$$

$\mathcal{B}$ runs in about the same time as $\mathcal{A}$.

Analysis:

We bound $\Pr[\text{CHALL}]$ via the OW-PCVA adversary $\mathcal{B}$.

Thanks to PTcheck, $\mathcal{B}$ no longer needs to guess which query triggered CHALL and gives m^* :

$\mathcal{B}$ looks for $(m', c^*, K') \in T_H$ that satisfy PTcheck(m', c^*).

$$\Pr[\text{CHALL}] \leq \text{Adv}_{\text{PKE}}^{\text{OW-PCVA}}(\mathcal{B}).$$

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Step 2: U-transform - PKE to IND-C_A KEM

We correct the PKE requirement in our security claim:

Security claim (PKE OW-PCVA $\Rightarrow$ KEM[⊥] IND-C_A)

For any IND-C_A adversary $\mathcal{A}$ against KEM[⊥], issuing at most q_H queries to random oracle H , there exists an OW-PCVA adversary $\mathcal{B}$ against PKE such that

$$\text{Adv}_{\text{KEM}^\perp}^{\text{IND-C}_A}(\mathcal{A}) \leq q_H \cdot \text{Adv}_{\text{PKE}}^{\text{OW-PCVA}}(\mathcal{B}).$$

$\mathcal{B}$ runs in about the same time as $\mathcal{A}$.

Key insights:

We used:

- **Lazy sampling** and **Unpredictability** as before.
- **“Programmable” ROM**: $\mathcal{B}$ programs RO responses to ensure consistency between Decaps and H .

For C_A, we need more from PKE than just OW.

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Step 2: U-transform - PKE to IND-CCA KEM

For CCA, we needed more from PKE than just OW.

That's annoying:

- OW-PCVA is stronger and less natural than OW.
- Fewer PKEs qualify, and proving OW-PCVA can be tedious.

Requiring only “plain” OW would mean:

- It works from the minimal standard notion.
- Applicability to many more PKEs, and the proofs are conceptually cleaner.

But how can we enable IND-CCA from OW? Two ways:

1. Augment U-transform in a way so that we don't need security in the presence of **Valid**.
2. Assume that PKE is deterministic and show that we don't need security in the presence of **PTcheck**.